

# Mosquito Lagoon Aquatic Preserve

## SEACAR Water Quality Analysis

Last compiled on 09 April, 2026

### Contents

|                                                                |          |
|----------------------------------------------------------------|----------|
| <b>Indicators</b>                                              | <b>2</b> |
| Nutrients . . . . .                                            | 2        |
| Total Nitrogen - Discrete . . . . .                            | 2        |
| Total Phosphorus - Discrete . . . . .                          | 4        |
| Water Quality . . . . .                                        | 6        |
| Dissolved Oxygen - Discrete . . . . .                          | 6        |
| Dissolved Oxygen Saturation - Discrete . . . . .               | 8        |
| Salinity - Discrete . . . . .                                  | 10       |
| Water Temperature - Discrete . . . . .                         | 12       |
| pH - Discrete . . . . .                                        | 14       |
| Water Clarity . . . . .                                        | 16       |
| Turbidity - Discrete . . . . .                                 | 16       |
| Total Suspended Solids - Discrete . . . . .                    | 18       |
| Chlorophyll a, Uncorrected for Pheophytin - Discrete . . . . . | 20       |
| Chlorophyll a, Corrected for Pheophytin - Discrete . . . . .   | 22       |
| Secchi Depth - Discrete . . . . .                              | 24       |
| Colored Dissolved Organic Matter - Discrete . . . . .          | 26       |

# Indicators

## Nutrients

### Total Nitrogen - Discrete

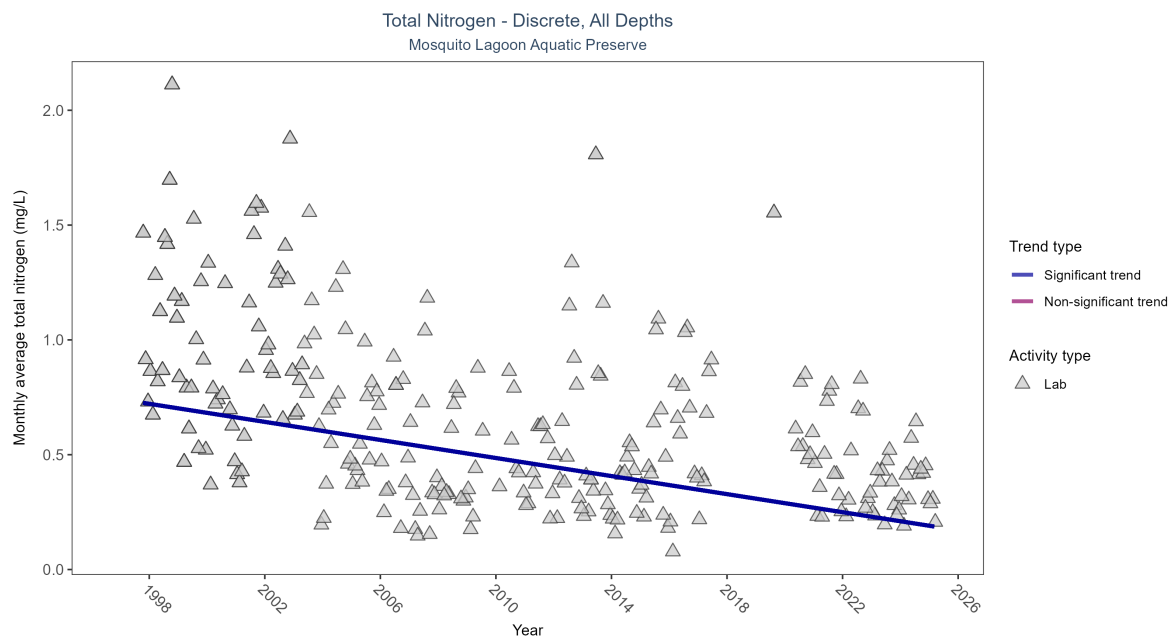


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Seasonal Kendall-Tau Results for - Total Nitrogen

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau      | Sen Intercept | Sen Slope | P |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|----------|---------------|-----------|---|
| Lab           | Significantly decreasing trend | 384          | 28              | 1997 - 2025      | 0.68912             | -0.42125 | 0.74129       | -0.01967  | 0 |

Monthly average total nitrogen decreased by 0.02 mg/L per year.

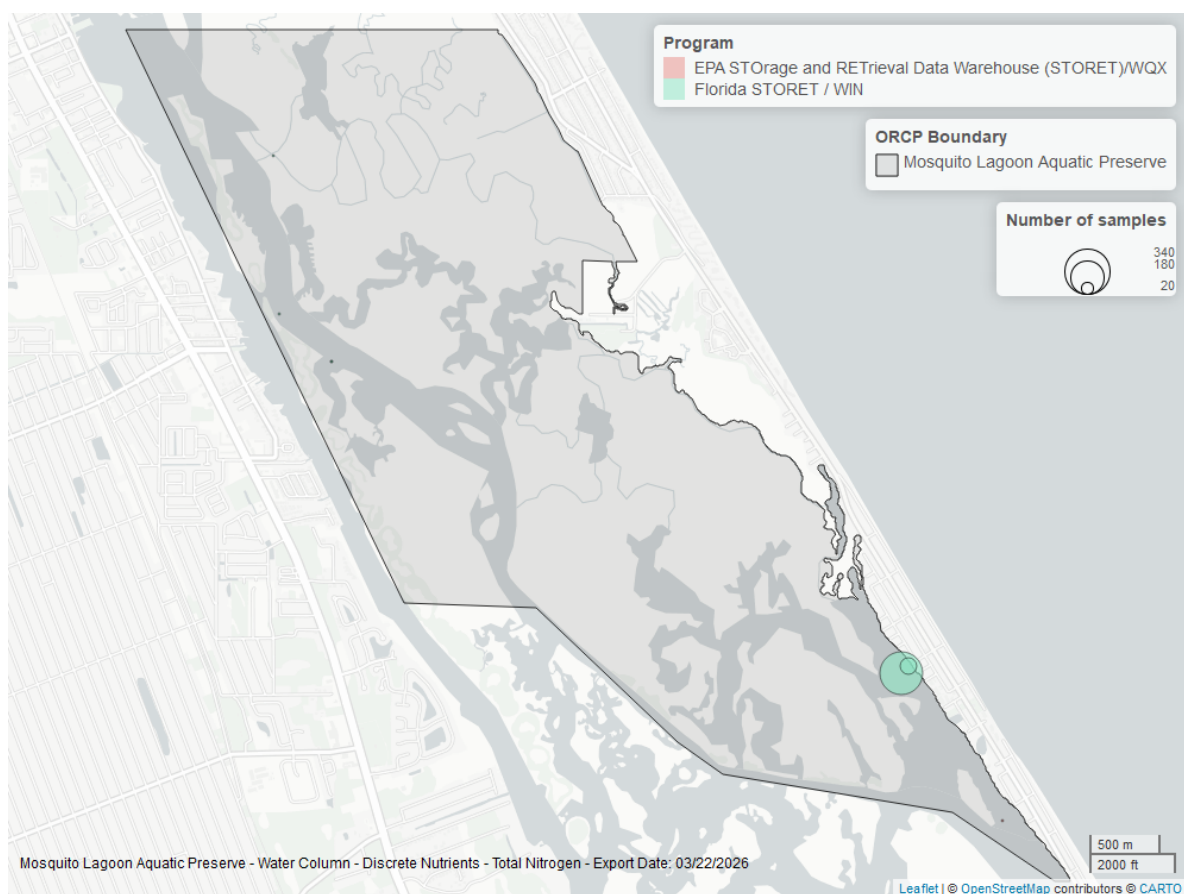


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Total Phosphorus - Discrete

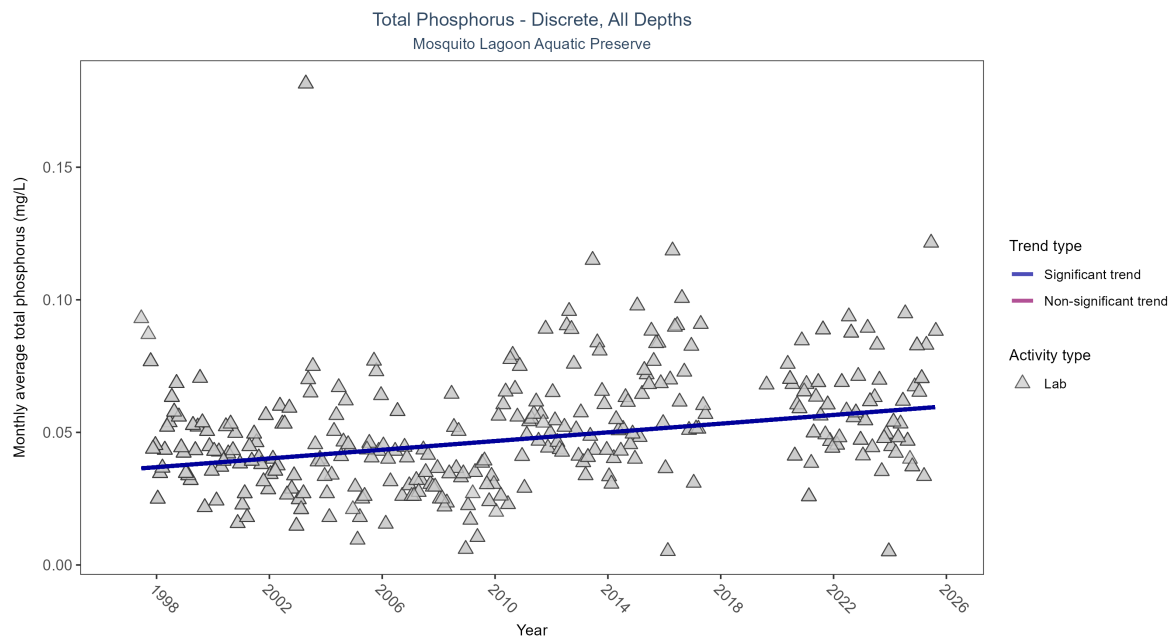


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Seasonal Kendall-Tau Results for - Total Phosphorus

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|---------|---------------|-----------|---|
| Lab           | Significantly increasing trend | 854          | 28              | 1997 - 2025      | 0.0422              | 0.26456 | 0.03606       | 0.00082   | 0 |

Monthly average total phosphorus increased by less than 0.01 mg/L per year.

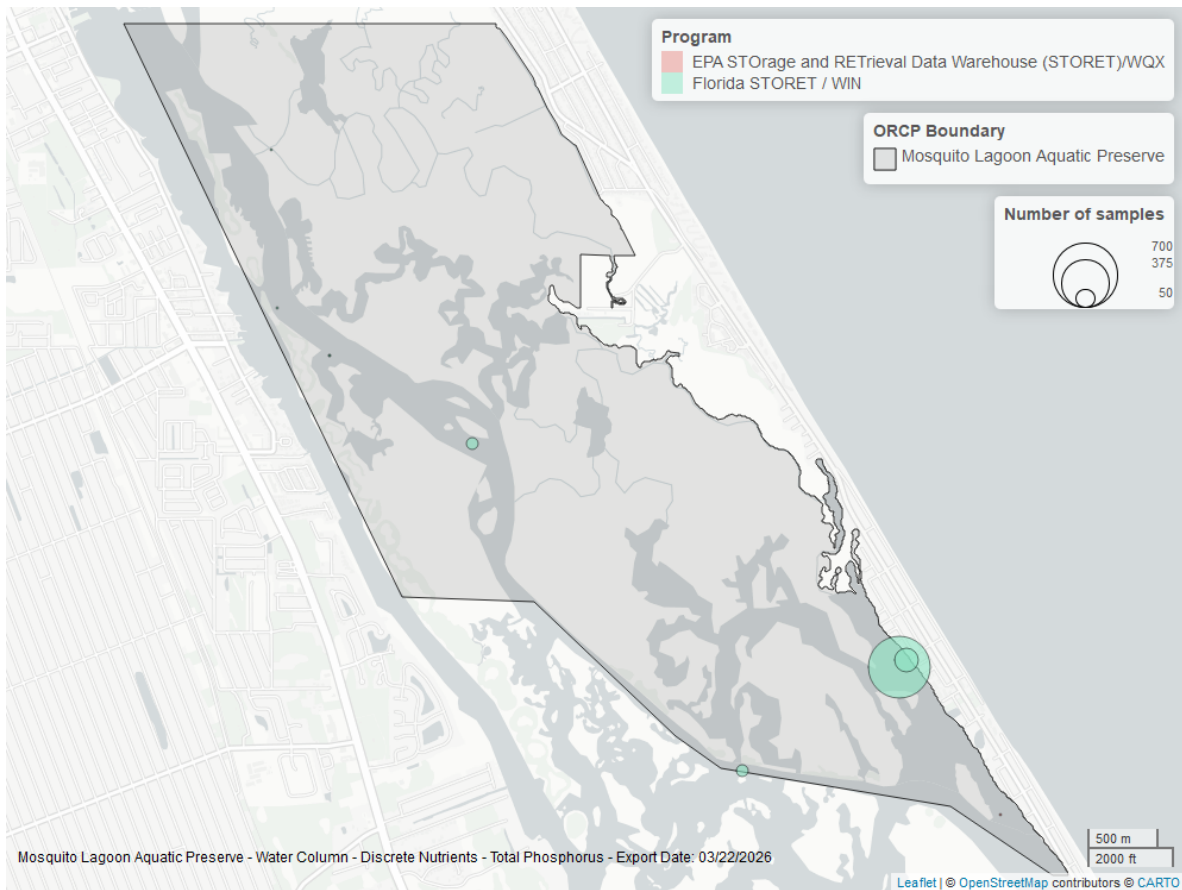


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Water Quality

### Dissolved Oxygen - Discrete

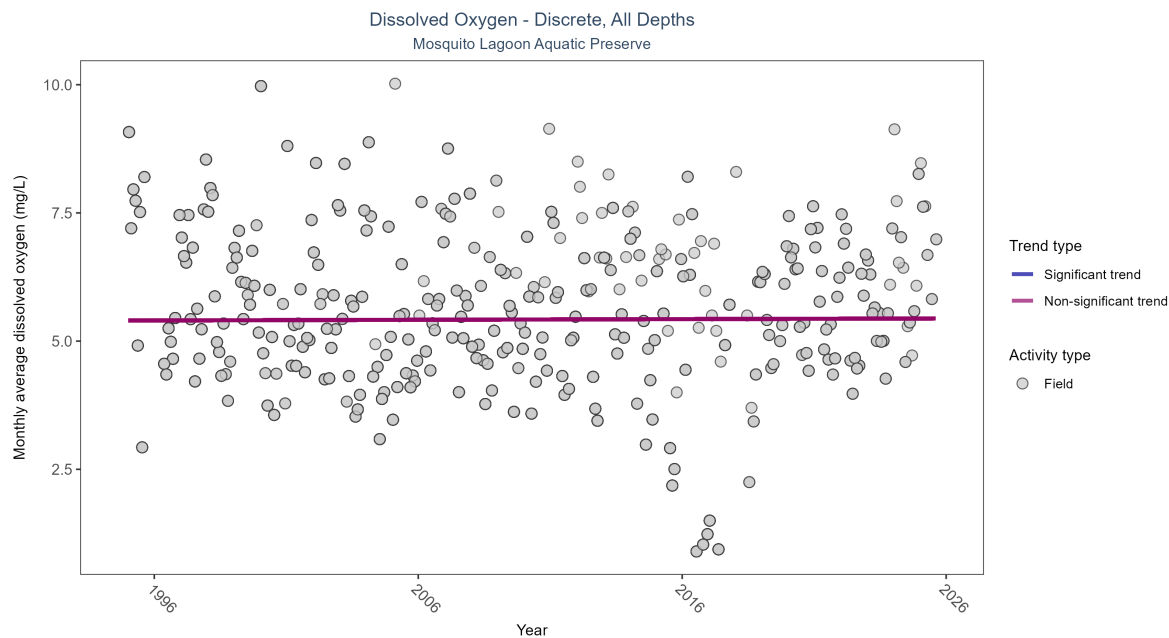


Figure 5: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 3: Seasonal Kendall-Tau Results for - Dissolved Oxygen

| Activity Type | Statistical Trend    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P     |
|---------------|----------------------|--------------|-----------------|------------------|---------------------|---------|---------------|-----------|-------|
| Field         | No significant trend | 4804         | 31              | 1995 - 2025      | 5.4                 | 0.00716 | 5.40124       | 0.00132   | 0.869 |

Dissolved oxygen showed no detectable trend between 1995 and 2025.

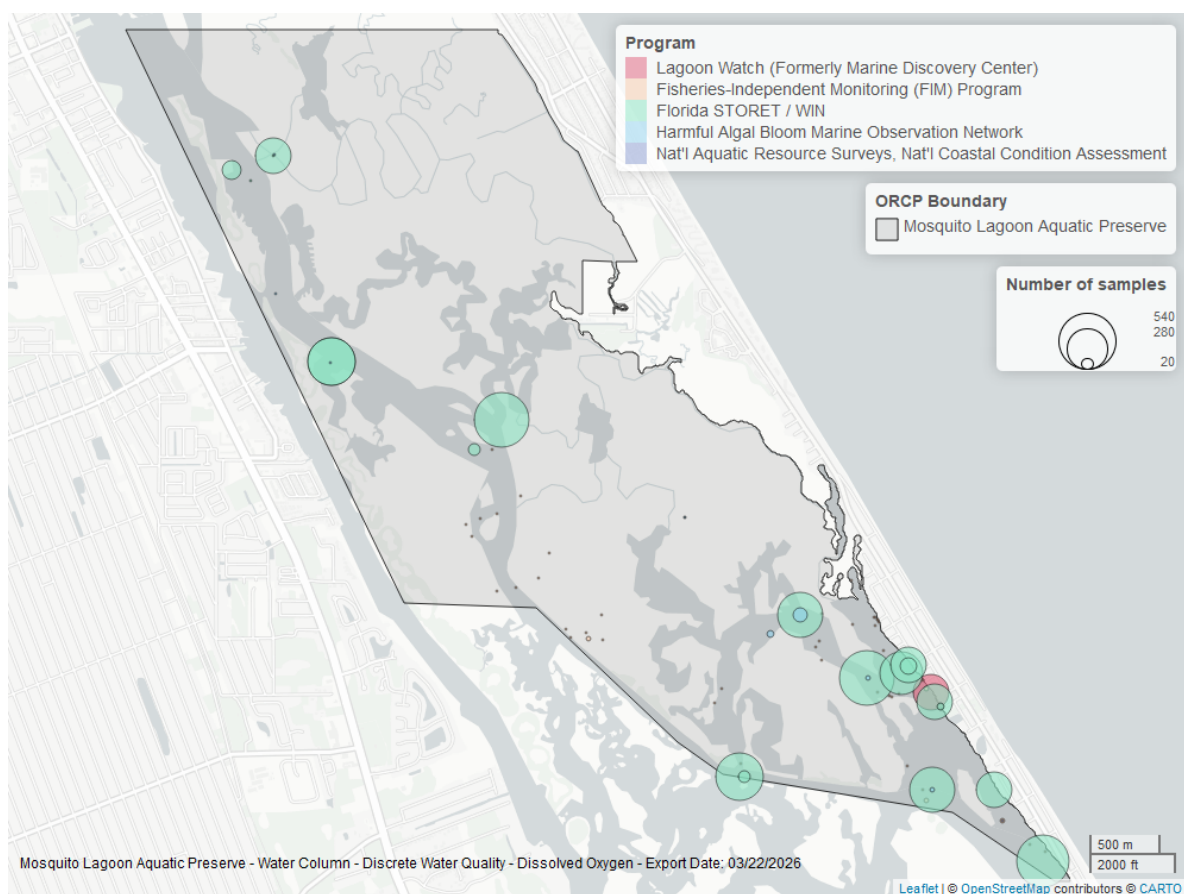


Figure 6: Map showing location of discrete water quality sampling locations within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Dissolved Oxygen Saturation - Discrete

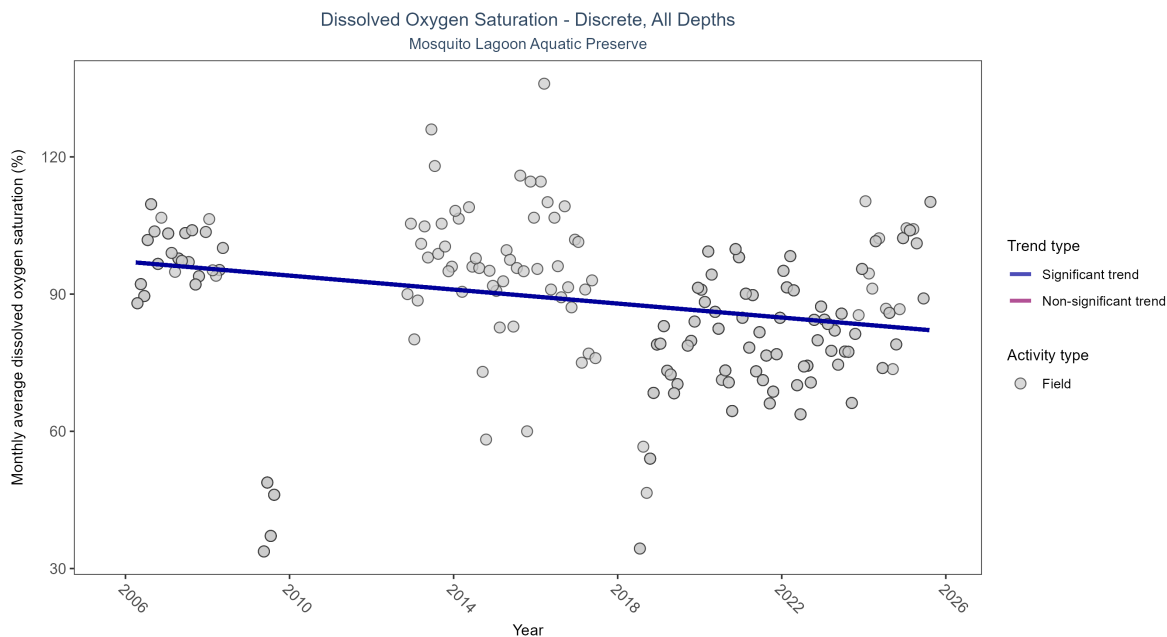


Figure 7: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 4: Seasonal Kendall-Tau Results for - Dissolved Oxygen Saturation

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau      | Sen Intercept | Sen Slope | P      |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|----------|---------------|-----------|--------|
| Field         | Significantly decreasing trend | 499          | 18              | 2006 - 2025      | 86.6                | -0.17844 | 97.10367      | -0.76429  | 0.0029 |

Monthly average dissolved oxygen saturation decreased by 0.76% per year.



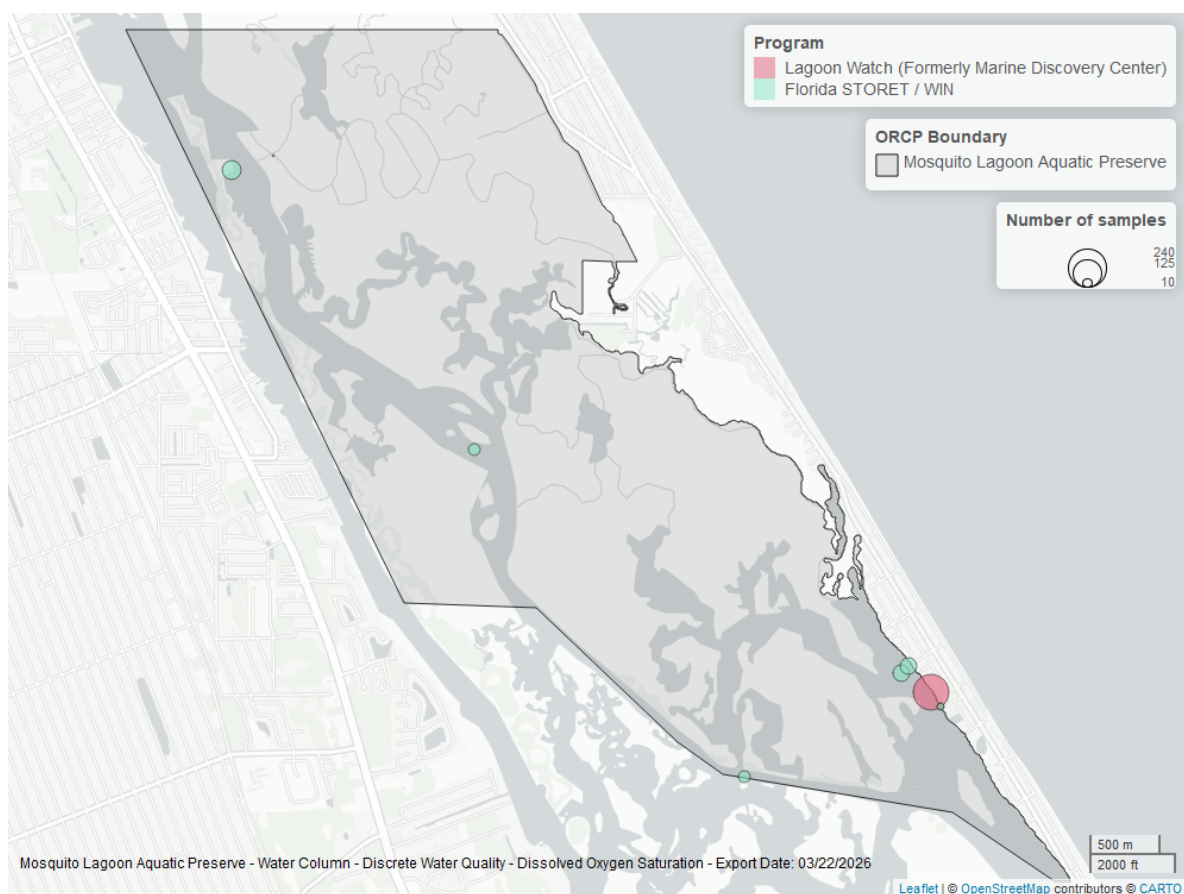


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Salinity - Discrete

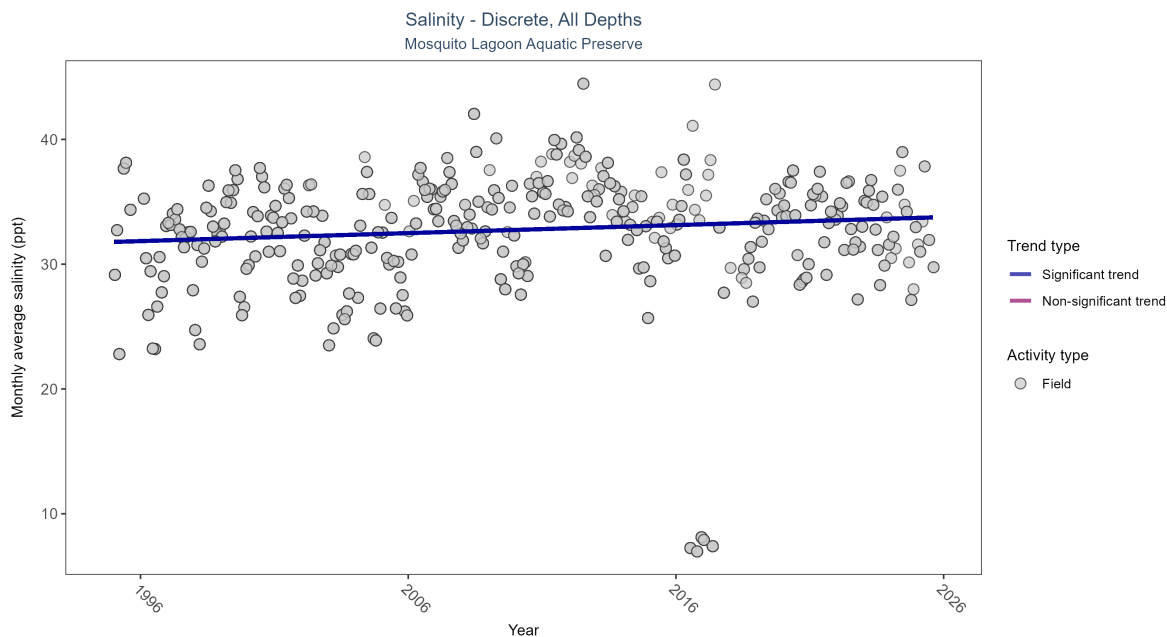


Figure 9: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 5: Seasonal Kendall-Tau Results for - Salinity

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P      |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|---------|---------------|-----------|--------|
| All           | Significantly increasing trend | 5262         | 31              | 1995 - 2025      | 32.5                | 0.11105 | 31.78411      | 0.06411   | 0.0034 |

Monthly average salinity increased by 0.06 ppt per year.

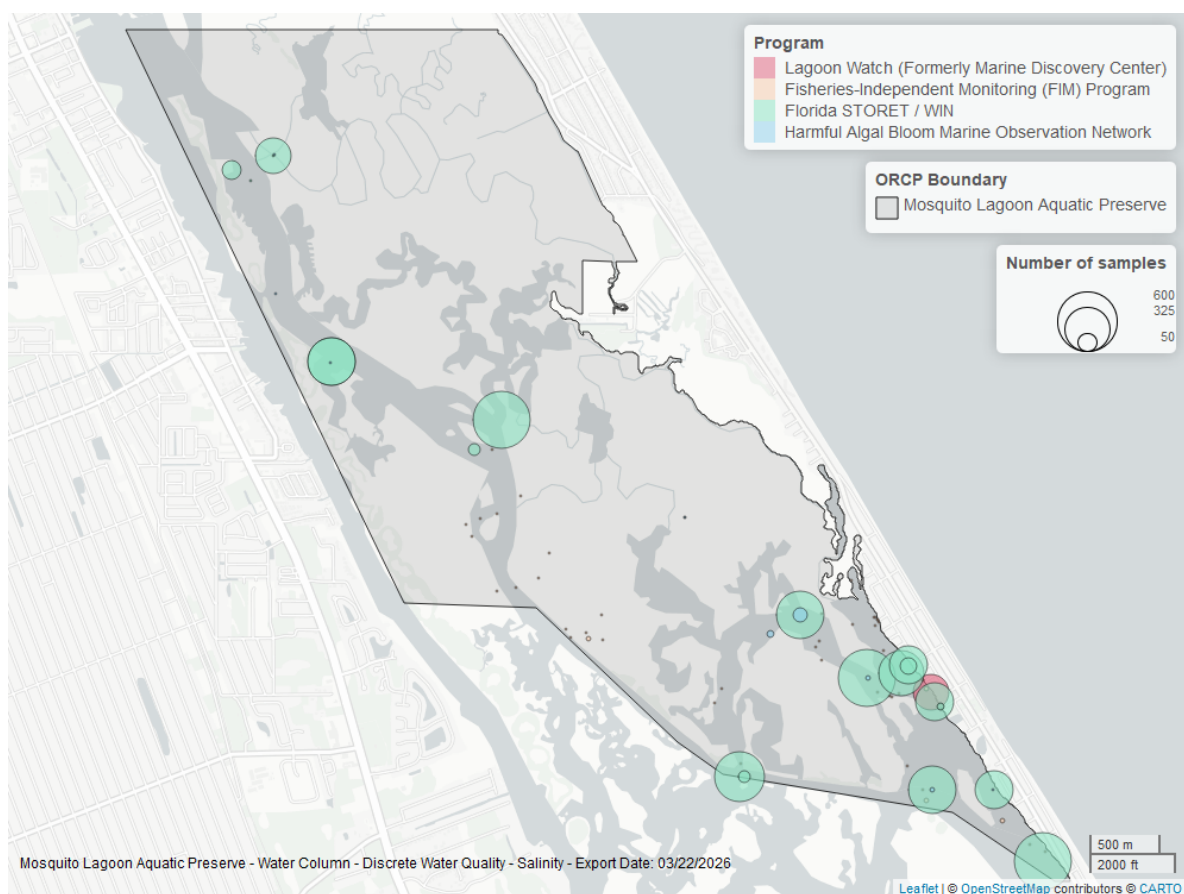


Figure 10: Map showing location of discrete water quality sampling locations within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Water Temperature - Discrete

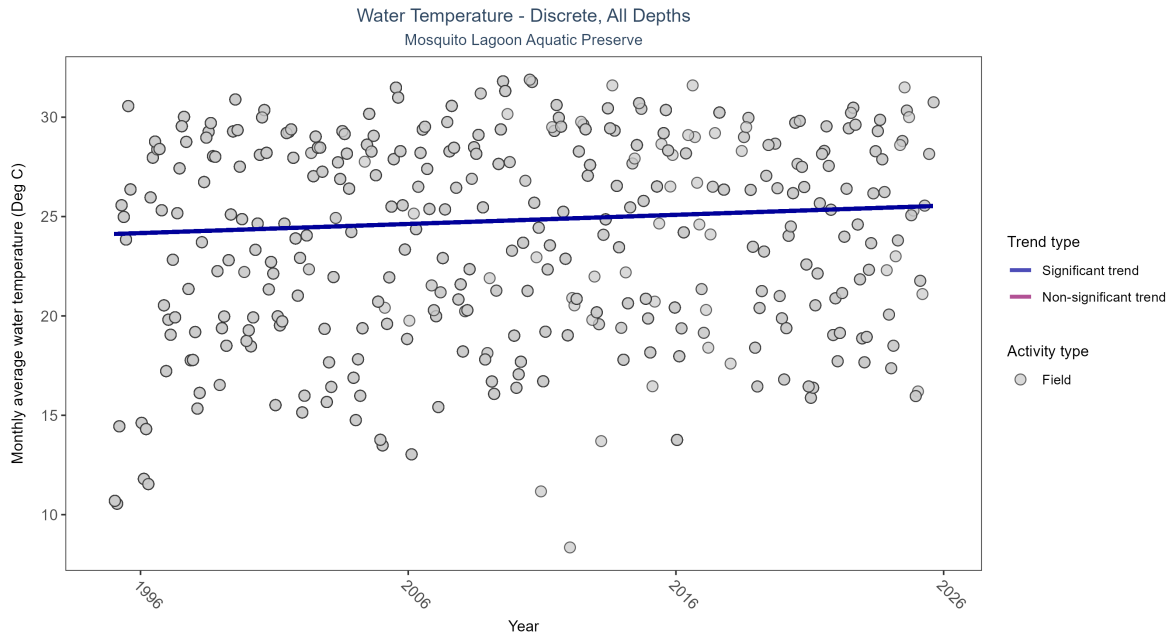


Figure 11: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 6: Seasonal Kendall-Tau Results for - Water Temperature

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|---------|---------------|-----------|---|
| Field         | Significantly increasing trend | 5351         | 31              | 1995 - 2025      | 25                  | 0.15935 | 24.1223       | 0.04595   | 0 |

Monthly average water temperature increased by 0.05°C per year.

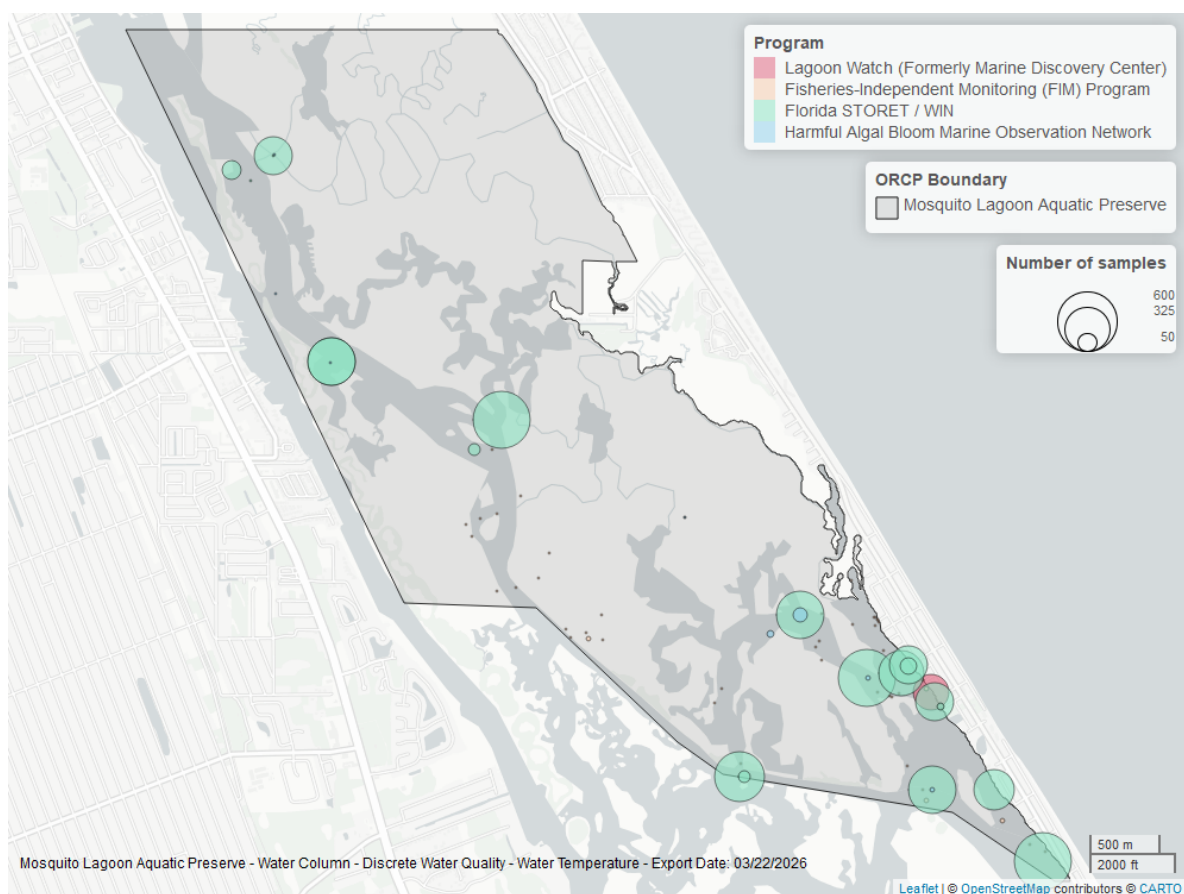


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## pH - Discrete

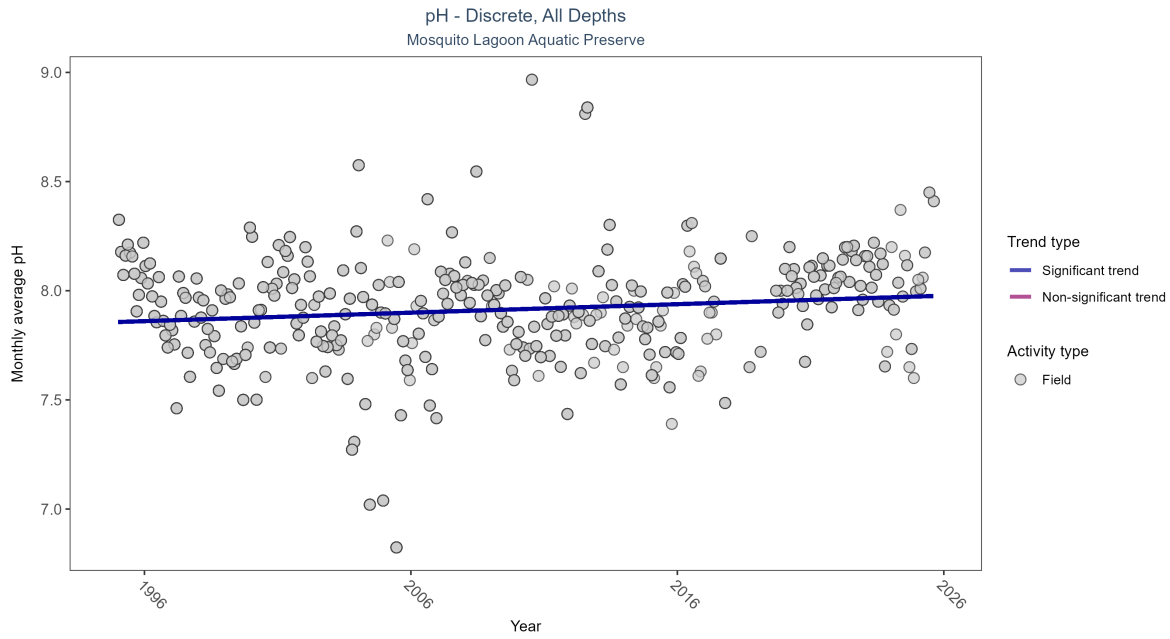


Figure 13: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 7: Seasonal Kendall-Tau Results for - pH

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P     |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|---------|---------------|-----------|-------|
| Field         | Significantly increasing trend | 3719         | 31              | 1995 - 2025      | 7.9                 | 0.11355 | 7.8567        | 0.00389   | 0.003 |

Monthly average pH increased by less than 0.01 pH units per year.

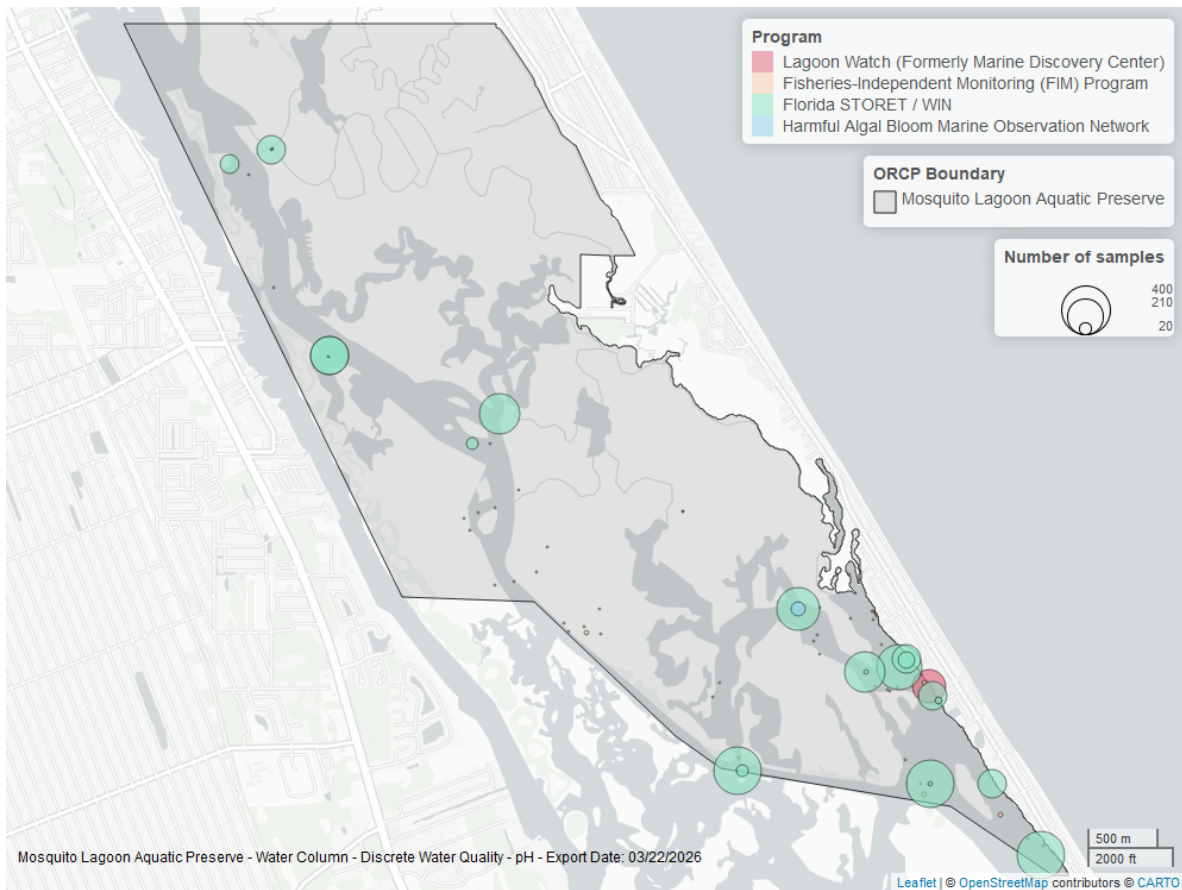


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Water Clarity

### Turbidity - Discrete

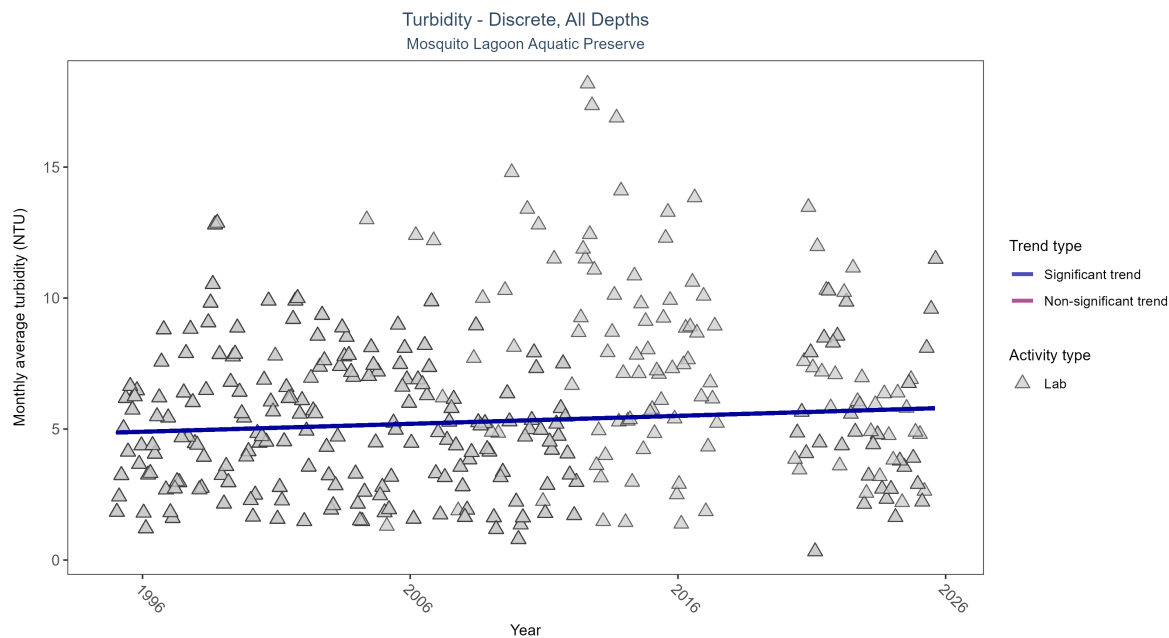


Figure 15: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 8: Seasonal Kendall-Tau Results for - Turbidity

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P      |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|---------|---------------|-----------|--------|
| Lab           | Significantly increasing trend | 3513         | 29              | 1995 - 2025      | 4.6                 | 0.09104 | 4.86621       | 0.03033   | 0.0174 |

Monthly average turbidity increased by 0.03 NTU per year, indicating a decrease in water clarity.



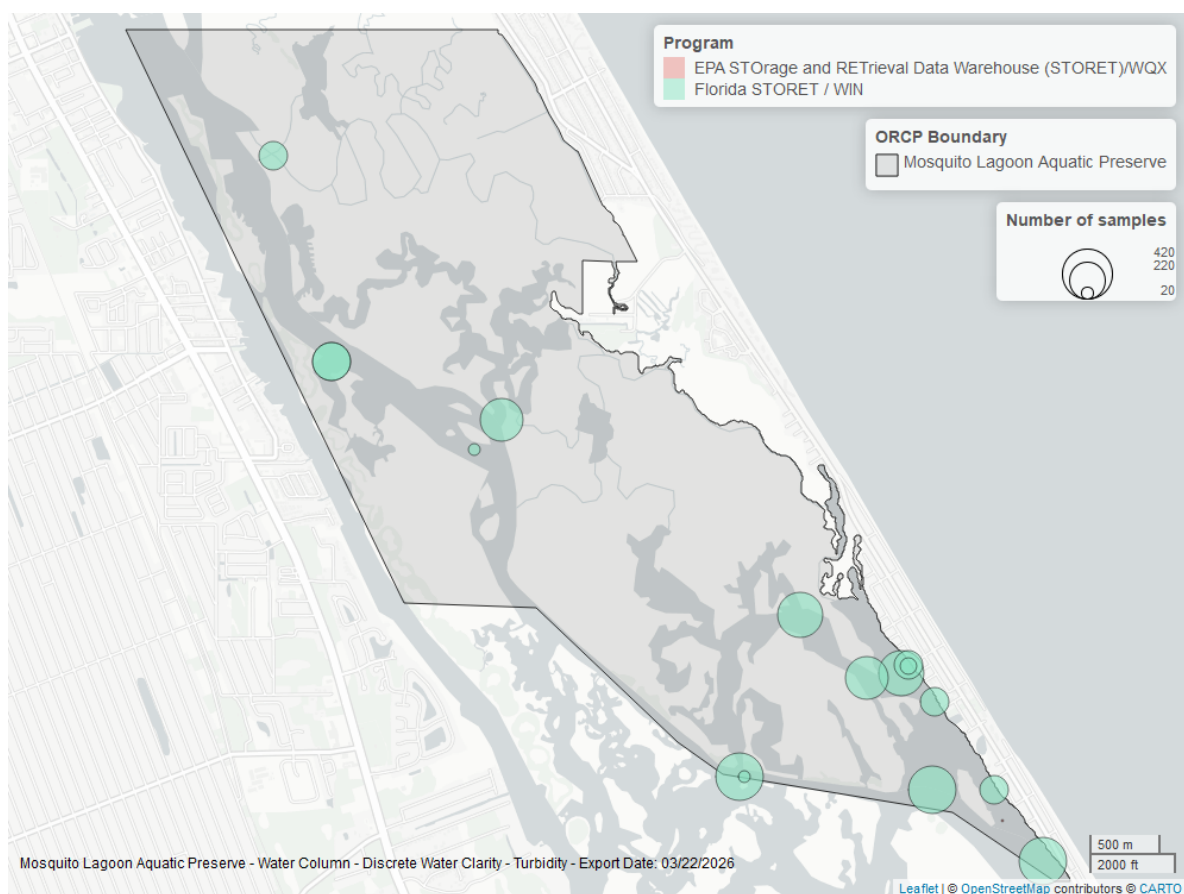


Figure 16: Map showing location of discrete water quality sampling locations within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Total Suspended Solids - Discrete

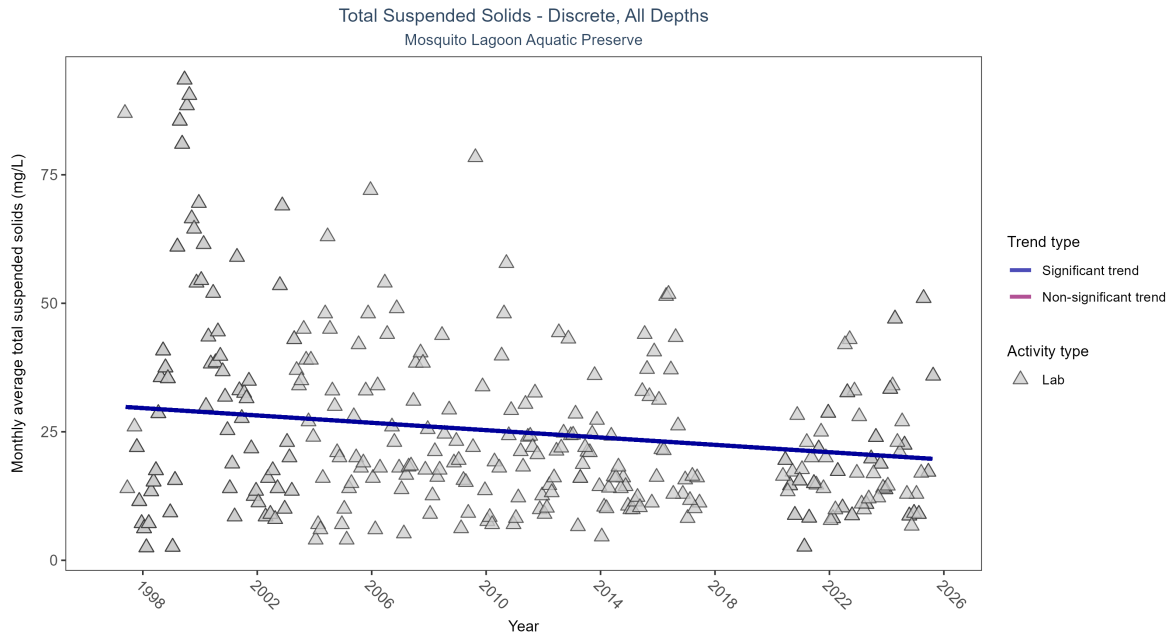


Figure 17: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 9: Seasonal Kendall-Tau Results for - Total Suspended Solids

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau      | Sen Intercept | Sen Slope | P      |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|----------|---------------|-----------|--------|
| Lab           | Significantly decreasing trend | 446          | 27              | 1997 - 2025      | 19.27495            | -0.17356 | 29.96786      | -0.35757  | 0.0001 |

Monthly average total suspended solids decreased by 0.36 mg/L per year, indicating an increase in water clarity.

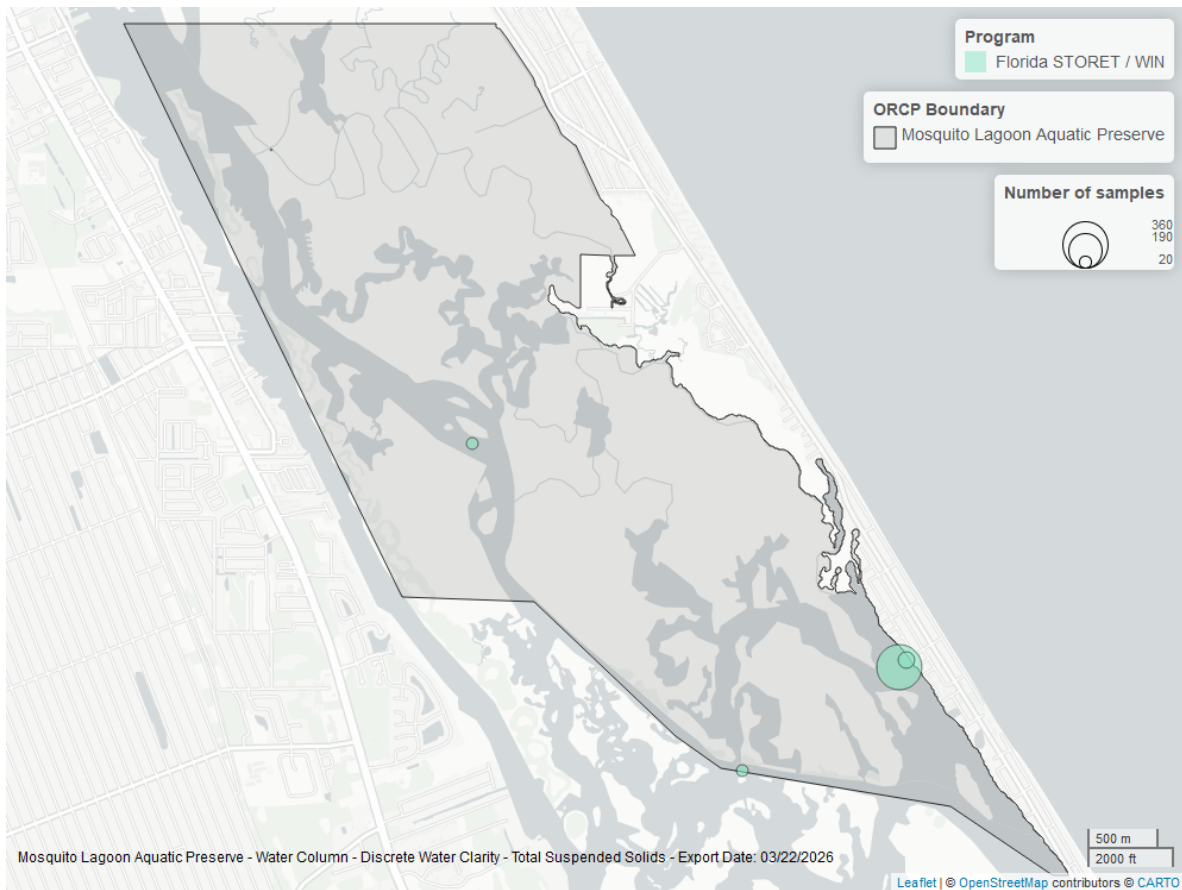


Figure 18: Map showing location of discrete water quality sampling locations within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Chlorophyll a, Uncorrected for Pheophytin - Discrete

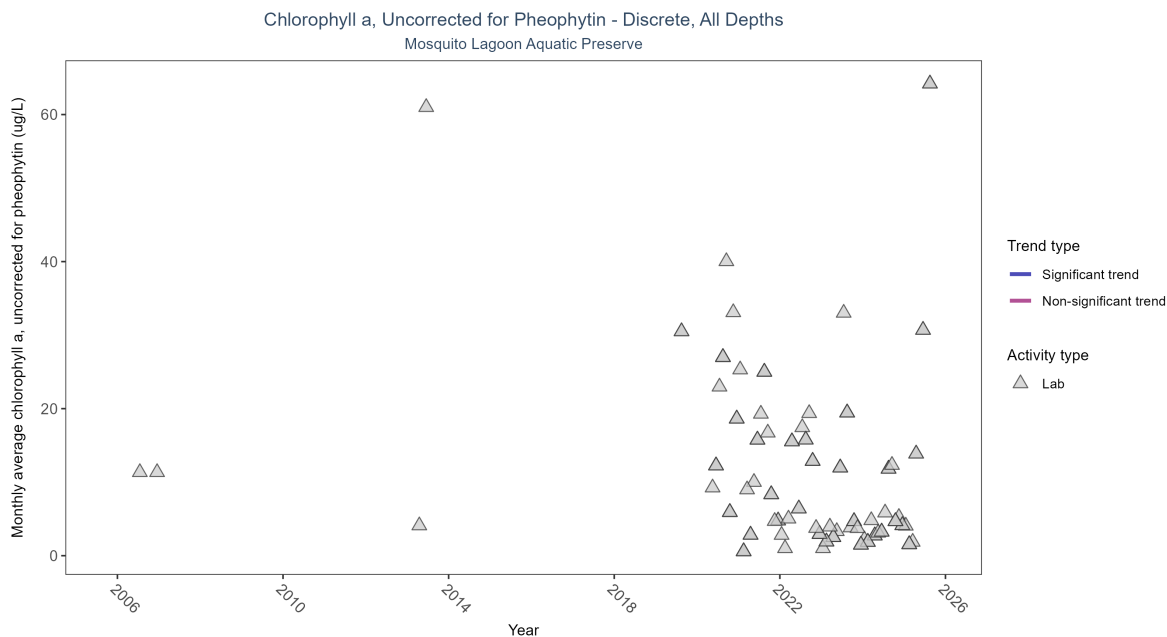


Figure 19: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 10: Seasonal Kendall-Tau Results for - Chlorophyll a, Uncorrected for Pheophytin

| Activity Type | Statistical Trend                    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---------------|--------------------------------------|--------------|-----------------|------------------|---------------------|-----|---------------|-----------|---|
| Lab           | Insufficient data to calculate trend | 124          | 9               | 2006 - 2025      | 5.8616              | -   | -             | -         | - |

There was insufficient data to fit a model for chlorophyll a, uncorrected for pheophytin.

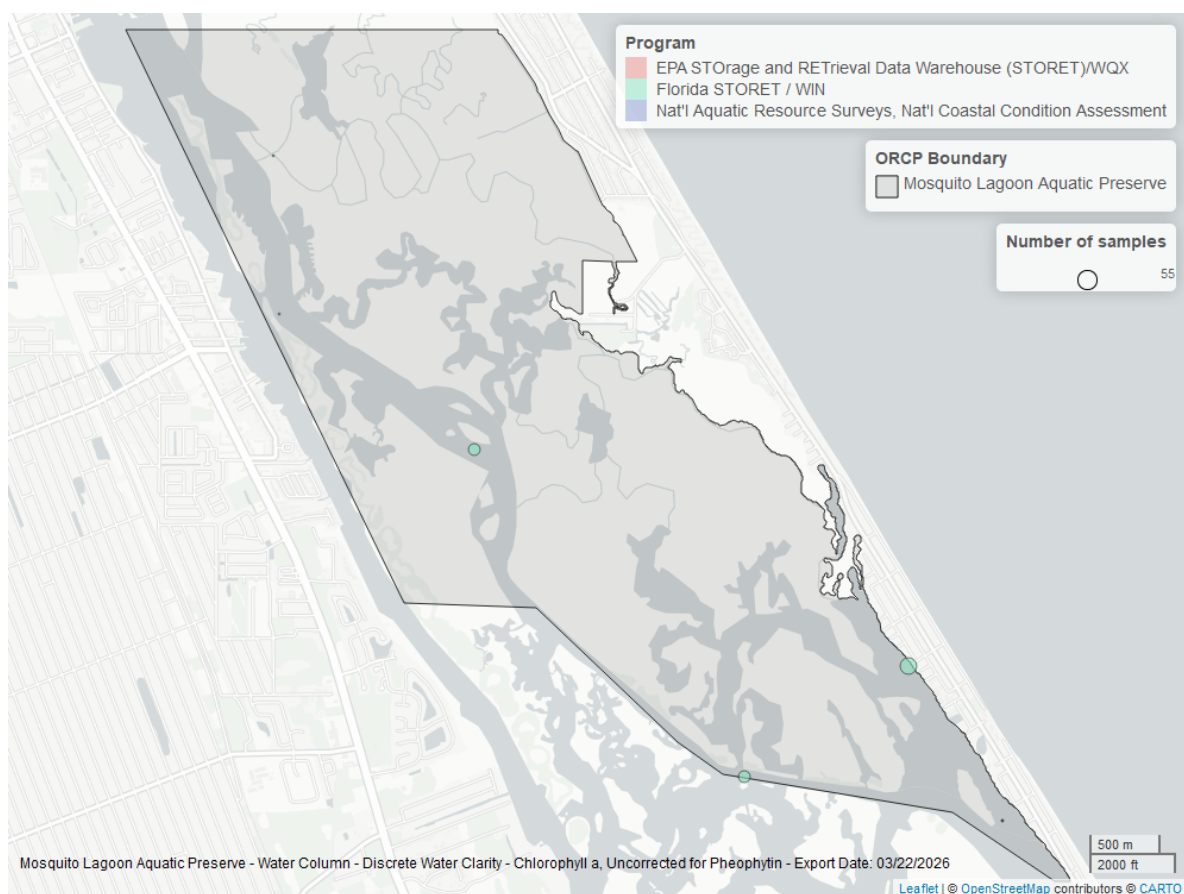


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Chlorophyll a, Corrected for Pheophytin - Discrete

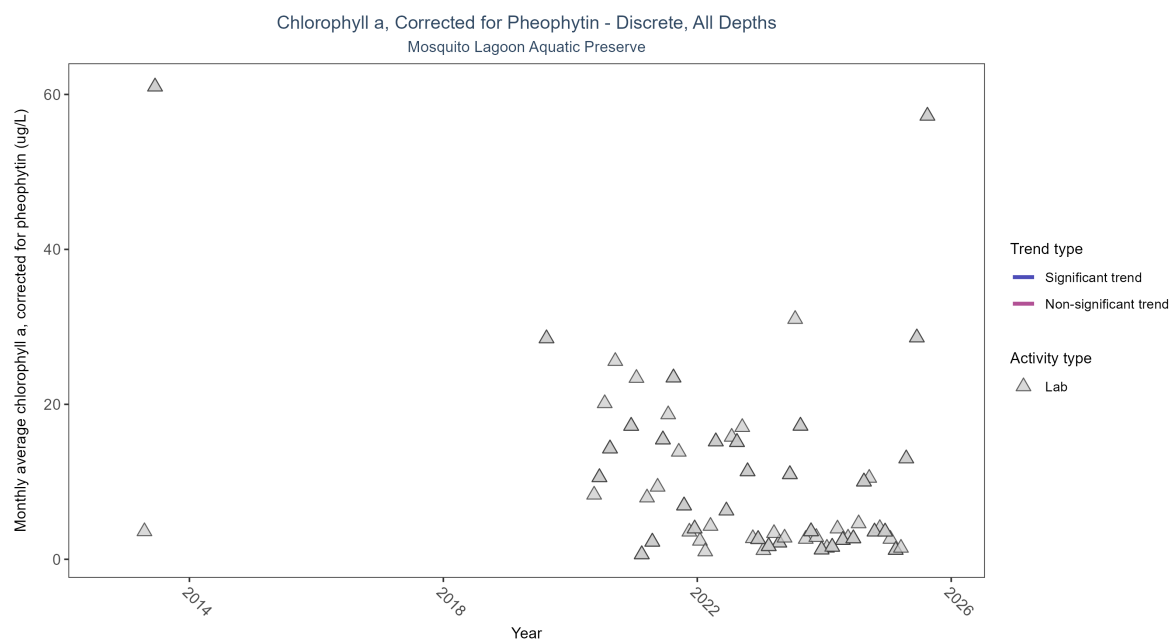


Figure 21: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 11: Seasonal Kendall-Tau Results for - Chlorophyll a, Corrected for Pheophytin

| Activity Type | Statistical Trend                    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---------------|--------------------------------------|--------------|-----------------|------------------|---------------------|-----|---------------|-----------|---|
| Lab           | Insufficient data to calculate trend | 119          | 8               | 2013 - 2025      | 4.9128              | -   | -             | -         | - |

There was insufficient data to fit a model for chlorophyll a, corrected for pheophytin.

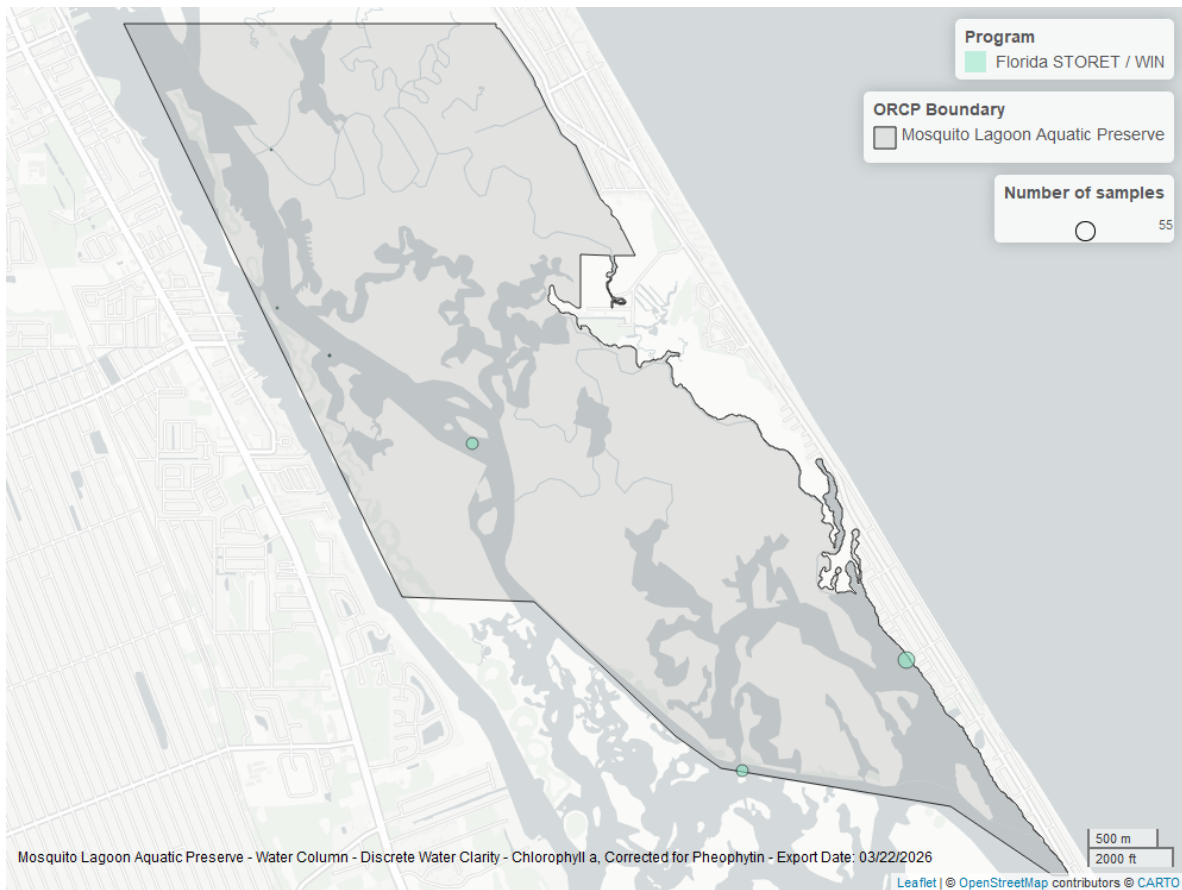


Figure 22: Map showing location of discrete water quality sampling locations within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Secchi Depth - Discrete

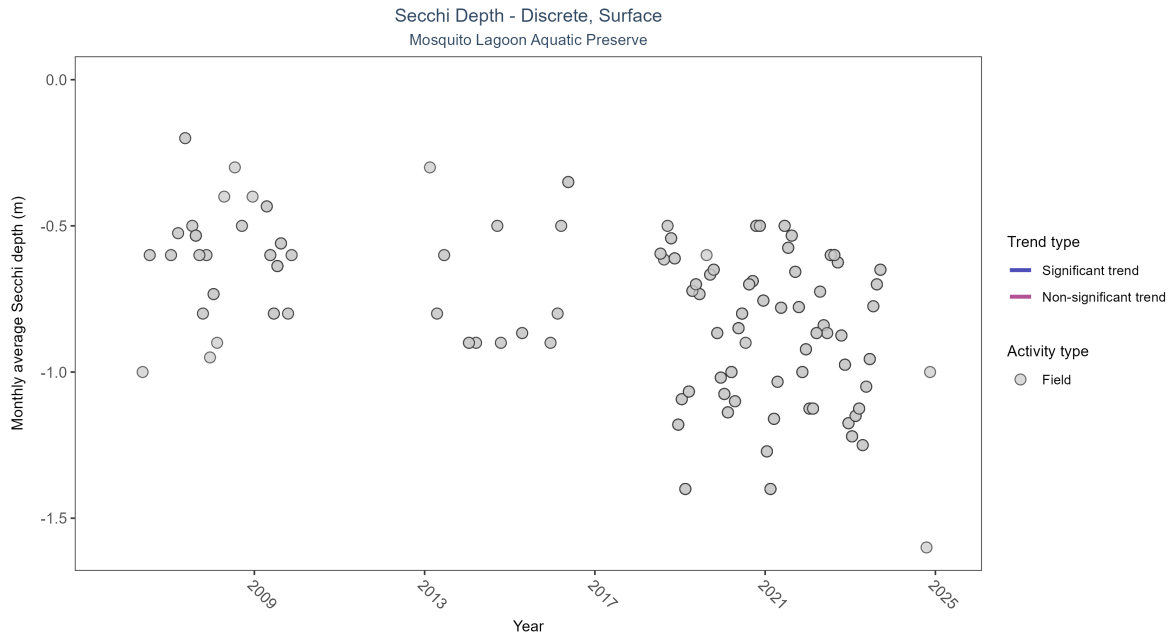


Figure 23: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 12: Seasonal Kendall-Tau Results for - Secchi Depth

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau      | Sen Intercept | Sen Slope | P |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|----------|---------------|-----------|---|
| Field         | Significantly decreasing trend | 688          | 19              | 2006 - 2025      | -0.8                | -0.46162 | -0.42681      | -0.02852  | 0 |

Monthly average Secchi depth became deeper by 0.03 m per year, indicating an increase in water clarity.



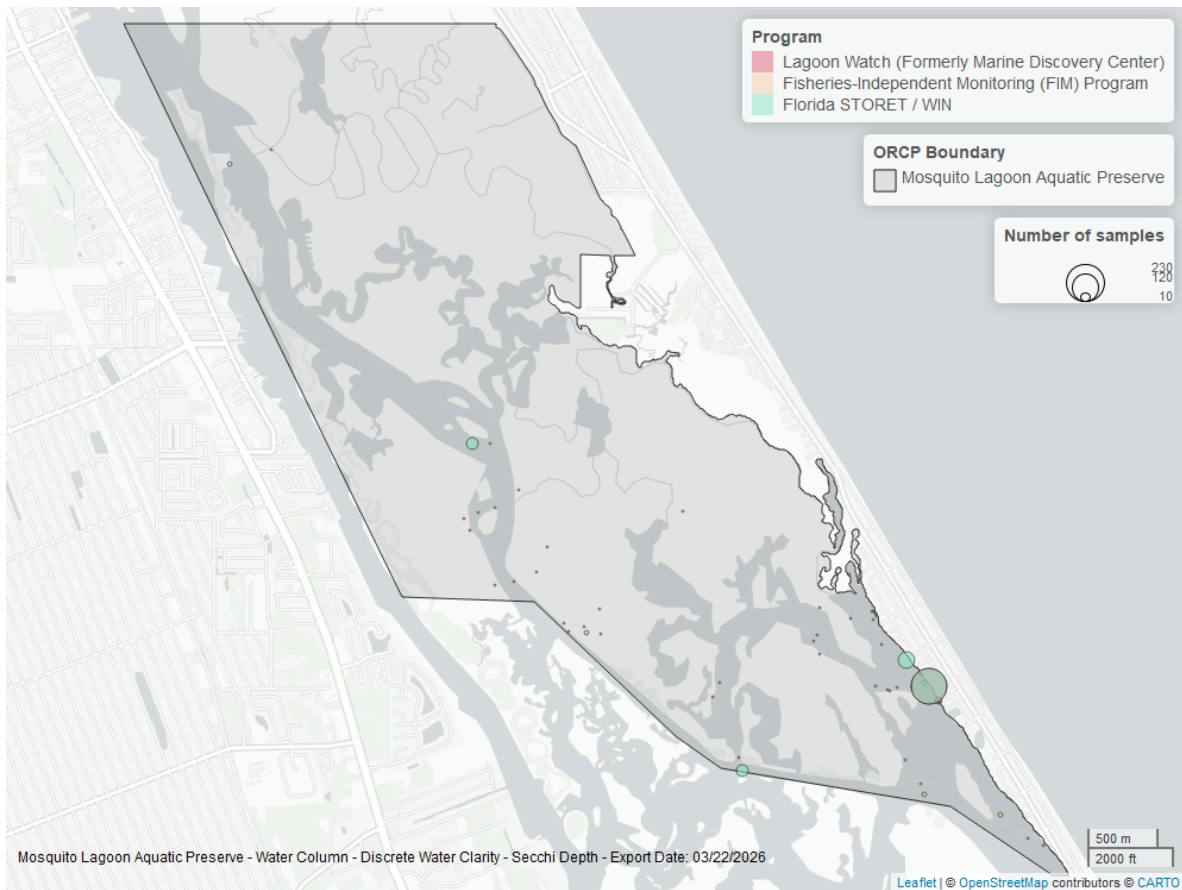


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

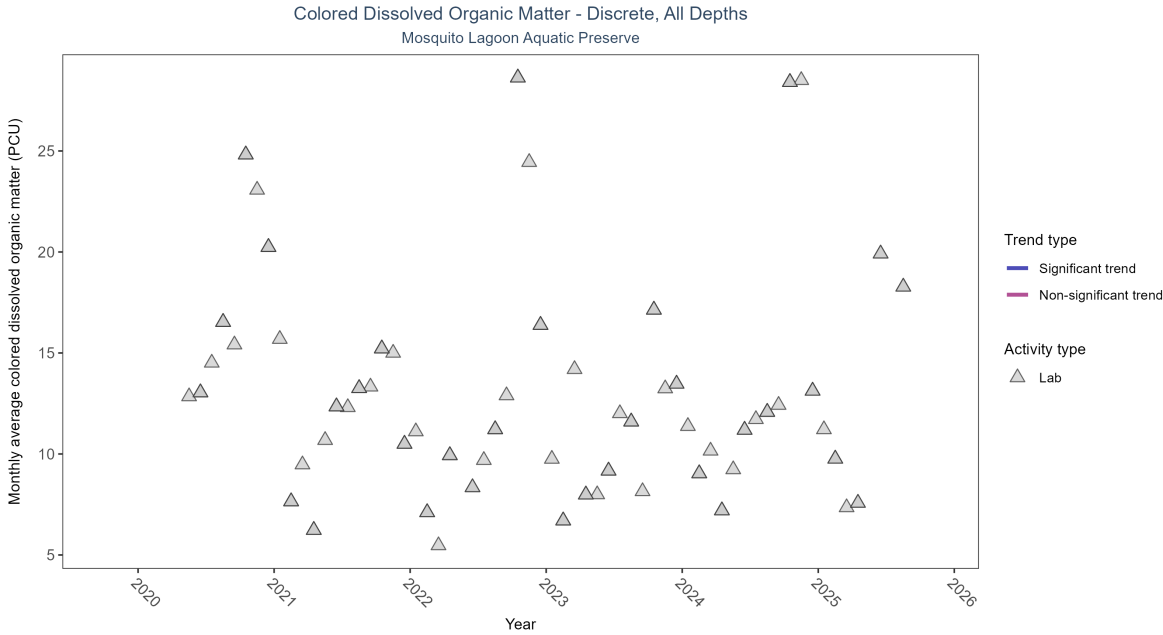


Figure 25: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 13: Seasonal Kendall-Tau Results for - Colored Dissolved Organic Matter

| Activity Type | Statistical Trend                    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---------------|--------------------------------------|--------------|-----------------|------------------|---------------------|-----|---------------|-----------|---|
| Lab           | Insufficient data to calculate trend | 121          | 6               | 2020 - 2025      | 11.73611            | -   | -             | -         | - |

There was insufficient data to fit a model for colored dissolved organic matter.

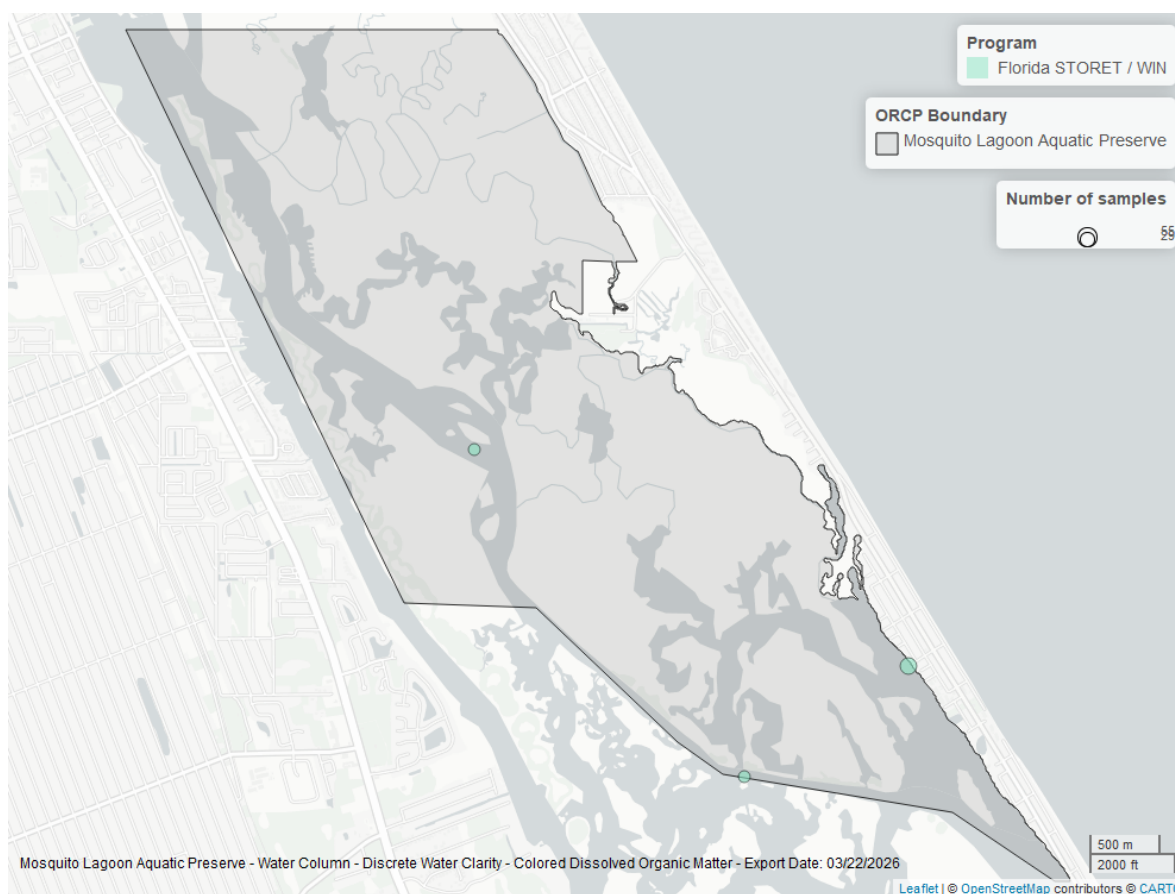


Figure 26: Map showing location of discrete water quality sampling locations within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.